

1 When $\text{Cov}(X, Y) = 0$, then X and Y are said to be **uncorrelated**.

2 Further, the **correlation between X and Y** is defined to be

3
$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}}$$

4 Properties that you should know:

5 1. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

6 2. $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$

7 3. $\text{Cov}(X, X) = V(X)$

8 4. $\text{Cov}(\underline{aX + b}, \underline{cY + d}) = ac \text{Cov}(X, Y)$

9 5. If X and Y are independent, then $\text{Cov}(X, Y) = 0$.

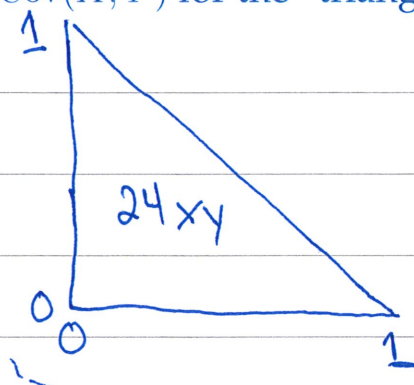
10 6. $-1 \leq \text{Corr}(X, Y) \leq 1$

11 **Exercise:** Is there a general condition under which X and Y uncorrelated
12 implies that X and Y are independent?

13 If (X, Y) is bivariate normal (to be
14 introduced soon)

15 Note: X, Y both normal $\nRightarrow (X, Y)$ is
16 bivariate normal. See example on
17 previous page.
18

Exercise: Find $\text{Cov}(X, Y)$ for the "triangular support" example.



Seems that $\text{Cov}(X, Y) < 0$

Recall: $X \sim \text{Beta}(2, 3)$, $Y \sim \text{Beta}(2, 3)$

$$E(X) = E(Y) = \frac{2}{5}$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x,y) dx dy$$

$$= \int_0^1 \int_0^{1-y} 24x^2 y^2 dx dy$$

$$= \frac{8}{60}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = -\frac{1}{15} = -\frac{2}{75}$$

Exercise: Suppose that X and Y are random variables with $Y = aX + b$ for constants a and b with $a \neq 0$. What is the correlation between X and Y ?

$$\begin{aligned}\text{Cov}(X, Y) &= \text{Cov}(X, aX + b) \\ &= a\text{Cov}(X) \quad \text{using prop 4} \\ &= aV(X)\end{aligned}$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}}$$

$$V(Y) = V(aX + b) = a^2 V(X)$$

$$= \frac{aV(X)}{\sqrt{a^2 V(X)V(X)}} = \frac{a}{|a|}$$

$$= \text{sgn}(a) = \begin{cases} 1, & a > 0 \\ -1, & a < 0 \end{cases}$$

The Bivariate Normal Distribution

The pair (X_1, X_2) has the **bivariate normal distribution** if it has joint pdf

$$f(\mathbf{x}) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

← vector 2x1
← determinant scalar

where $\boldsymbol{\mu}$ is the vector of means and Σ is the **covariance matrix**:

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_1, X_2) & \text{Var}(X_2) \end{bmatrix}$$

Some important properties, when (X_1, X_2) bivariate normal:

1. X_1 and X_2 are normally distributed, i.e. the marginals are normal.
2. The conditional distribution of X_1 given $X_2 = x_2$ is normal with mean

$$\mu_1 + \frac{\sigma_1}{\sigma_2} \rho (x_2 - \mu_2)$$

and variance

$$(1 - \rho^2) \sigma_1^2$$

where ρ equals the correlation between X_1 and X_2 .



3. All linear combinations of X_1 and X_2 are also normally distributed.
4. X_1 and X_2 uncorrelated implies that X_1 and X_2 are independent.

Also: X_1 and X_2 normal and independent implies that (X_1, X_2) is bivariate normal

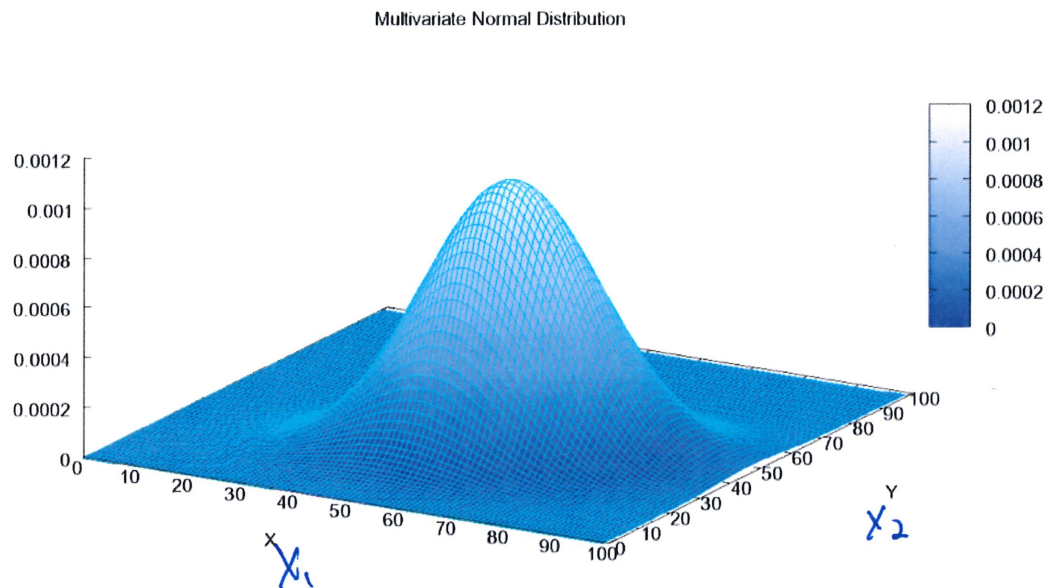
In general,

$$Z = aX_1 + bX_2$$

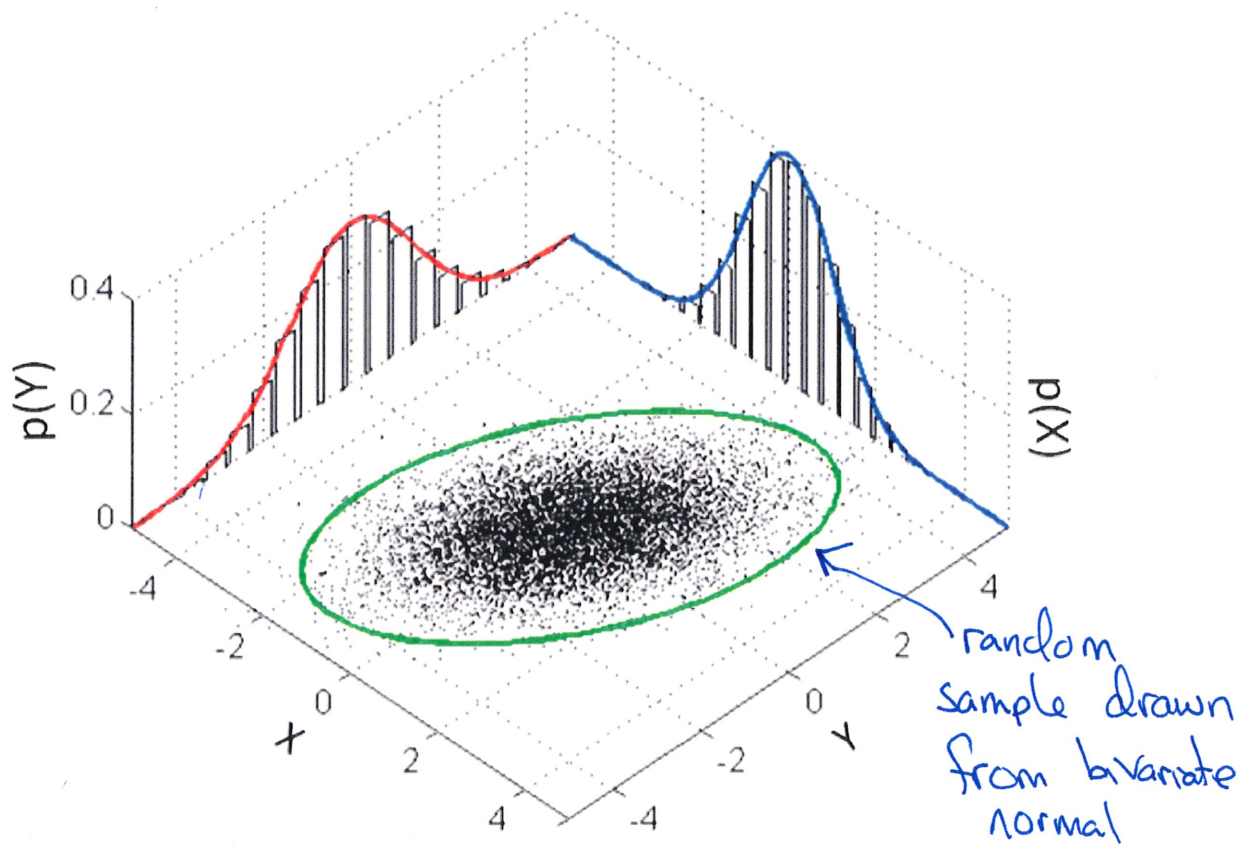
$$E(Z) = aE(X_1) + bE(X_2)$$

$$V(Z) = V(aX_1) + V(bX_2) + 2\text{Cov}(aX_1, bX_2)$$

$$= a^2 V(X_1) + b^2 V(X_2) + \underbrace{2ab \text{Cov}(X_1, X_2)}_{=0 \text{ if } X_1, X_2 \text{ uncorrelated}}$$



1



2

3 Figures from Wikipedia.

More than Two Random Variables

As one moves to the joint distribution of more than two random variables, it becomes increasingly important to work in matrix notation.

Define a **random vector** of length n by writing

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$$

where each of the X_i is a random variable. Then we can define

$$E(\mathbf{X}) = \begin{bmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_n) \end{bmatrix}$$

and $V(\mathbf{X})$ is a n by n matrix whose (i, j) element is $\text{Cov}(X_i, X_j)$. (Of course, this means that the diagonal holds the variances of the X_i .)

It is standard to let $\underline{\mu} = E(\mathbf{X})$ and $\underline{\Sigma} = V(\mathbf{X})$.

$$\underline{\Sigma} = \begin{bmatrix} V(X_1) & \text{Cov}(X_1, X_2) & \cdots \\ & V(X_2) & \\ & & \ddots \\ \text{Cov}(X_n, X_1) & \cdots & V(X_n) \end{bmatrix}$$

Some important properties: Let $\mathbf{u}$ and $\mathbf{v}$ denote two vectors of n scalars.

$$1. E(\mathbf{u}^T \mathbf{X}) = \mathbf{u}^T E(\mathbf{X}) = \mathbf{u}^T \boldsymbol{\mu} \quad \left[\begin{array}{c} \phantom{\mathbf{u}^T \boldsymbol{\mu}} \\ \phantom{\mathbf{u}^T \boldsymbol{\mu}} \\ \phantom{\mathbf{u}^T \boldsymbol{\mu}} \end{array} \right]$$

$$2. V(\mathbf{u}^T \mathbf{X}) = \mathbf{u}^T \boldsymbol{\Sigma} \mathbf{u}$$

$$3. \text{Cov}(\mathbf{u}^T \mathbf{X}, \mathbf{v}^T \mathbf{X}) = \mathbf{u}^T \boldsymbol{\Sigma} \mathbf{v}$$

$$4. \text{ If } \mathbf{A} \text{ is } p \text{ by } n \text{ (non-random) matrix, then } E(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\mu} \text{ and } V(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T$$

$\mathbf{A}\mathbf{X} \sim p\text{-dim. vector}$

Exercise: Of course, the variance of a random variable cannot be negative.

What is an analogous constraint that is placed on $\boldsymbol{\Sigma}$?

$\boldsymbol{\Sigma}$ must be positive semidefinite
(i.e. $\mathbf{u}^T \boldsymbol{\Sigma} \mathbf{u} \geq 0$ for all $\mathbf{u}$)

$\boldsymbol{\Sigma}$ is positive definite if $\mathbf{u}^T \boldsymbol{\Sigma} \mathbf{u} > 0$
for any $\mathbf{u} \neq 0$

$\boldsymbol{\Sigma}$ p.d. $\Rightarrow \boldsymbol{\Sigma}^{-1}$ exists

Note: If $\boldsymbol{\Sigma}$ is p.s.d. then
 $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^T$ is also p.s.d.

Exercise: Suppose that X_1 , X_2 , and X_3 are random variables such that

$$E(X_1) = -1, E(X_2) = 0, E(X_3) = 2$$

and

$$V(X_1) = 1, V(X_2) = 3, V(X_3) = 1$$

and

$$\text{Cov}(X_1, X_2) = 0.5, \text{Cov}(X_1, X_3) = 0, \text{Cov}(X_2, X_3) = -0.5.$$

Let

$$Y_1 = X_1 - X_2 \leftarrow$$

and

$$Y_2 = X_1 + 2X_2 + 9X_3.$$

What are

$$E(Y_1), E(Y_2), V(Y_1), V(Y_2), \text{ and } \text{Cov}(Y_1, Y_2)?$$

Messy!

Exercise: Let $\mathbf{X}$ be a random vector such that

$$E(\mathbf{X}) = \boldsymbol{\mu} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

and

$$V(\mathbf{X}) = \boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0.5 & 0 \\ 0.5 & 3 & -0.5 \\ 0 & -0.5 & 1 \end{bmatrix}.$$

Let

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 9 \end{bmatrix}$$

What are the mean and variance of $\mathbf{Y} = \mathbf{A}\mathbf{X}$?

Same as on previous page.

$$E(\mathbf{Y}) = E(\mathbf{A}\mathbf{X}) = \mathbf{A} E(\mathbf{X}) = \mathbf{A} \boldsymbol{\mu}$$

$$= \begin{bmatrix} (1)(-1) + (-1)(0) + (0)(2) \\ (1)(-1) + (2)(0) + (9)(2) \end{bmatrix} = \begin{bmatrix} -1 \\ 17 \end{bmatrix}$$

$$V(\mathbf{Y}) = V(\mathbf{A}\mathbf{X}) = \mathbf{A} V(\mathbf{X}) \mathbf{A}^T = \mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^T = \begin{bmatrix} 3 & 0 \\ 0 & 78 \end{bmatrix}$$

$$\text{So, } E(Y_1) = -1, E(Y_2) = 17, V(Y_1) = 3, V(Y_2) = 78$$

$$\text{Cov}(Y_1, Y_2) = 0$$

Y_1, Y_2 are uncorrelated, but cannot say they are independent

1 Joint Distributions

2 We can write $f_{\mathbf{X}}(\mathbf{x})$ for the **joint distribution** of $\mathbf{X}$, much as we did in the
 3 bivariate case. 

4 For instance, if $\mathbf{X}$ is continuous,

$$5 \quad P(X_1 \in (a_1, b_1), X_2 \in (a_2, b_2), \dots, X_n \in (a_n, b_n)) =$$

$$6 \quad \int_{a_1}^{b_1} \int_{a_2}^{b_2} \cdots \int_{a_n}^{b_n} f_{\mathbf{X}}(\mathbf{x}) dx_n dx_{n-1} \cdots dx_1$$

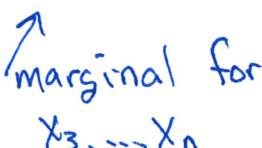
7 We obtain multivariate marginal and conditional distributions as before:

8 **Integrate out** to get marginals. For example,

$$9 \quad f_{(X_3, X_4, \dots, X_n)}(x_3, x_4, \dots, x_n) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\mathbf{X}}(\mathbf{x}) dx_1 dx_2$$

10 For the conditionals **divide by density** for the random variables being con-
 11 ditioned on. For example,

$$12 \quad f_{(X_1, X_2) | (X_3, X_4, \dots, X_n)}(\underline{x_1, x_2} | \underline{x_3, x_4, \dots, x_n}) = \frac{f_{\mathbf{X}}(\mathbf{x})}{f_{(X_3, X_4, \dots, X_n)}(x_3, x_4, \dots, x_n)}$$



 marginal for
 $x_3, \dots, x_n$

Exercise: Suppose that $X_1, X_2, \dots, X_n$ have joint pdf

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} \frac{2n!}{(2 + \sum_{i=1}^n x_i)^{n+1}}, & \text{if } x_i > 0, i = 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

For this exercise, assume that $n = 5$.

1. What is the marginal distribution for (X_1, X_4) ?

2. What is the conditional distribution of (X_2, X_3, X_5) given (X_1, X_4) ?

$$\textcircled{1} f_{(X_1, X_4)}(x_1, x_4) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\mathbf{X}}(\underline{x}) dx_2 dx_3 dx_5$$

$$= \int_0^{\infty} \int_0^{\infty} \int_0^{\infty} \frac{2n!}{(2 + \sum x_i)^{n+1}} dx_2 dx_3 dx_5, \quad x_1 > 0, x_4 > 0$$

$$= \frac{4}{(2 + x_1 + x_4)^3}, \quad x_1 > 0, x_4 > 0$$

$$\textcircled{2} \cancel{f_{(X_1, X_3, X_5)}(x_1, x_3, x_5)}$$

$$f_{(X_2, X_3, X_5 | X_1, X_4)}(x_2, x_3, x_5 | x_1, x_4) = \frac{f_{\mathbf{X}}(\underline{x})}{f_{(X_1, X_4)}(x_1, x_4)}$$

$$= \left(\frac{240}{(2 + \sum x_i)^6} \right) / \left(\frac{4}{(2 + x_1 + x_4)^3} \right), \quad x_i > 0$$

1 Independence

2 The random variables $X_1, X_2, \dots, X_n$ are independent if

$$3 \quad f_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^n f_{X_i}(x_i) \quad \text{for all } x_1, x_2, \dots, x_n$$

4 When say that “ $X_1, X_2, \dots, X_n$ are i.i.d.” that means that they are **inde-**
5 **pendent and identically distributed**. The term “identically distributed”
6 means that the random variables have the exact same distribution, not sim-
7 ply that they are “all normal” or “all exponential.”

1 Multivariate Normal

2 The vector $\mathbf{X}$ (of length n) has the **multivariate normal distribution** if it
3 has joint pdf

$$4 \quad f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

5 It is assumed that Σ is positive definite.

6 This implies that $E(\mathbf{X}) = \boldsymbol{\mu}$ and $V(\mathbf{X}) = \Sigma$. All marginal and condi-
7 tional distributions are multivariate normal. Further, any random vari-
8 able/vector of the form $\mathbf{AX}$ will also be multivariate normal, provided
9 that $\mathbf{A}\Sigma\mathbf{A}^T$ is positive definite.

10 If $\text{Cov}(X_i, X_j) = 0$, then X_i and X_j are independent, and if Σ is diagonal,
11 then $X_1, X_2, \dots, X_n$ are independent.

Exercise: Suppose that X is multivariate normal with

$$\mu_X = \begin{bmatrix} 4 \\ 6 \\ 3 \end{bmatrix}$$

and

$$\Sigma_X = \begin{bmatrix} 1 & 0.5 & 0 \\ 0.5 & 2 & 0.1 \\ 0 & 0.1 & 3 \end{bmatrix}.$$

- 1 What is the distribution for (X_1, X_3) ? Which pairs of random variables are
2 independent, if any?

3 (X_1, X_3) is bivariate normal with $\mu_Y = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$
4 and $\Sigma_Y = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$
5

6 We see that $\text{Cov}(X_1, X_3) = 0$, so X_1, X_3
7 are independent since they are bivariate normal
8

9 But: X_1, X_2 dependent
10 X_2, X_3 dependent
11
12
13
14

Example: Portfolio Theory

Let R_i denote the return on equity i for $i = 1, 2, \dots, n$, and let w_i denote the weight placed on equity i in the portfolio. Assume that $\sum_i w_i = 1$ and $w_i \geq 0$.

Let the overall return be $Y = \sum_i w_i R_i = \underline{w}^T \underline{R}$

Exercise: Describe the distribution of Y , and what properties an optimal portfolio would possess.

$$E(Y) = \underline{w}^T E(\underline{R}) \quad \text{want this to be large}$$

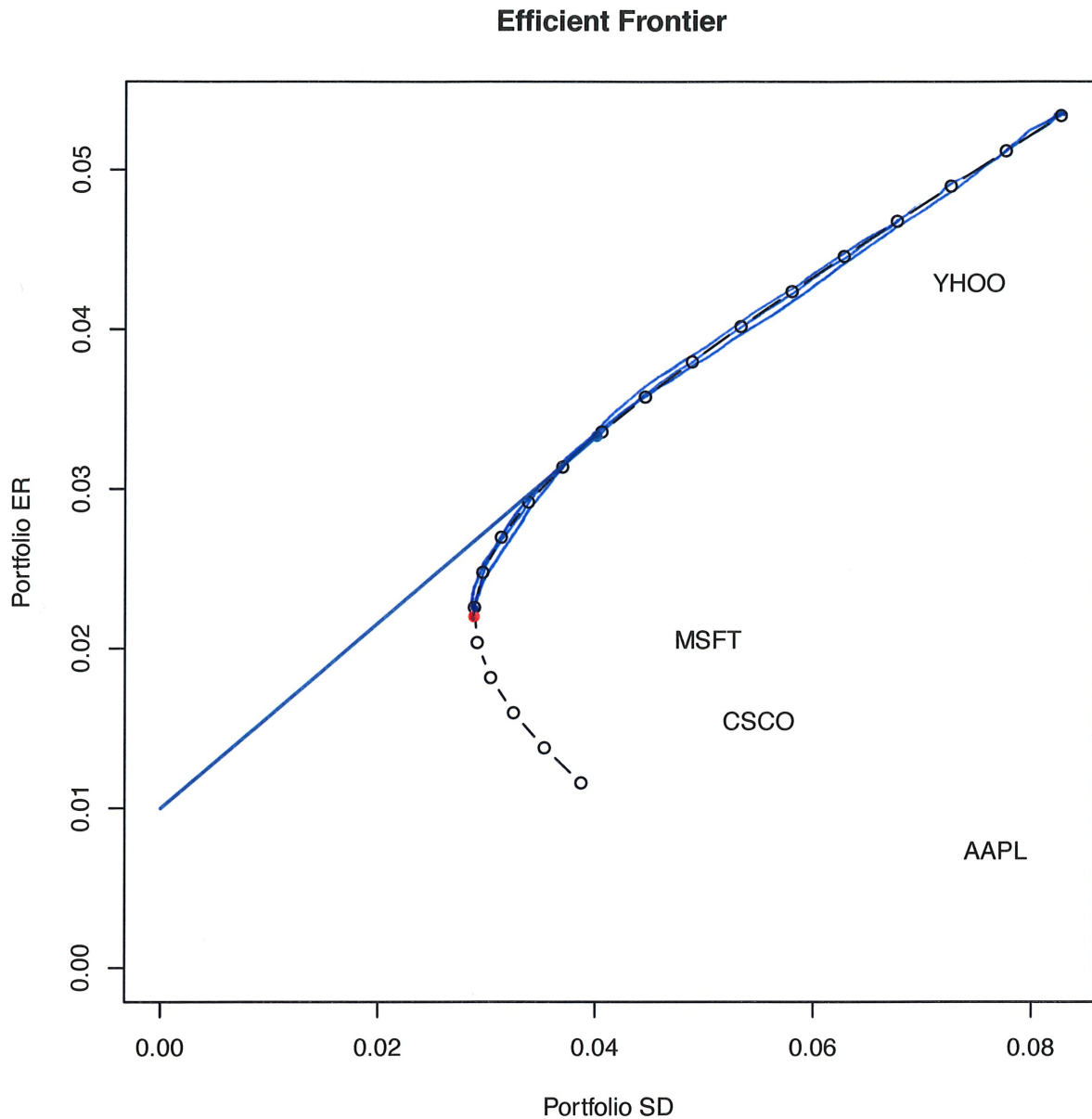
$$V(Y) = \underline{w}^T V(\underline{R}) \underline{w} \quad \text{want this to be small}$$

Any choice of $\underline{w}$ lands somewhere in the (SD, EV) plane

Among strategies with same EV, prefer one with smallest SD

Among those with same SD, prefer one with largest EV

This curves out the efficient frontier



1

2 Based on monthly returns from September 1, 2012 to September 1, 2014.

Part 4: Conditional Expectation

In this Part we will present some useful results concerning **conditional expectation**.

The simplest example of conditional expectation would likely be the case of a discrete random variable X , when conditioning on an event B , i.e.,

$$E(X|B) = \sum_k kP(X = k|B).$$

It is often the case that the event B is an event concerning X itself, e.g.,

$$E(X|\underbrace{X \geq k}).$$

The conditional distributions we derive in Part 3 play a key role here. We can naturally define

$$E(Y|X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy \quad \text{continuous}$$

or

$$E(Y|X = x) = \sum_y y f_{Y|X}(y|x) \quad \text{discrete}$$

in the cases where Y is continuous and discrete, respectively.

Exercise: Suppose that the joint pdf of (X, Y) is given by:

$$f_{XY}(x, y) = \begin{cases} 24xy & x > 0, y > 0 \\ & x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

What is the expected value of Y given that X equals x ?

$$f_{Y|X}(y|x) = \begin{cases} \frac{24xy}{12x(1-x)^2} & 0 < x < 1 \\ & 0 \leq y < 1-x \\ 0 & \text{otherwise} \end{cases}$$

$$E(Y|X=x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy$$

$$= \int_0^{1-x} \frac{2y^2}{(1-x)^2} dy, \quad 0 < x < 1$$

$$= \frac{2}{3} (1-x), \quad 0 < x < 1$$

$$E(Y) = \frac{2}{5} \neq E(Y|X=x)$$

$$E(Y) = \int_{-\infty}^{\infty} E(Y|X=x) f_X(x) dx = E(E(Y|X))$$

Laws of Total Probability

Assume that events $B_1, B_2, \dots$ form a partition of Ω , and that A is another event. Then each of the following hold.

$$P(A) = \sum_i P(A|B_i)P(B_i)$$

$$E(X) = \sum_i E(X|B_i)P(B_i)$$

$$P(A|C) = \sum_i P(A|C, B_i)P(B_i|C) \quad \leftarrow$$

$$E(X|C) = \sum_i E(X|C, B_i)P(B_i|C) \quad \leftarrow \text{used on next page}$$

$$f_X(x) = \sum_y f_{X|Y}(x|y) f_Y(y) \quad \text{if } Y \text{ discrete}$$

$$f_X(x) = \int \underline{f_{X|Y}(x|y)} f_Y(y) dy \quad \text{if } Y \text{ continuous}$$

Exercise (From HOTS): What is the expected number of tosses of an unfair coin needed to get two heads in a row (assume probability p of a head)?

X = # of tosses needed to see two heads in a row

Y_i = outcome of i^{th} toss (H, T)

$$\begin{aligned} E(X) &= E(X | Y_1 = H) P(Y_1 = H) + E(X | Y_1 = T) P(Y_1 = T) \\ &= \underbrace{E(X | Y_1 = H)}_{} p + (1 + E(X)) (1 - p) \end{aligned}$$

$$\begin{aligned} E(X | Y_1 = H) &= E(X | Y_1 = H, Y_2 = H) P(Y_2 = H | Y_1 = H) \\ &\quad + E(X | Y_1 = H, Y_2 = T) P(Y_2 = T | Y_1 = H) \\ &= (2)(p) + (2 + E(X))(1 - p) \end{aligned}$$

$$= E(X)(1 - p^2) + p + 1$$

Solve for $E(X)$, find $E(X) = \frac{p+1}{p^2}$